

Vamsi Krishna Kommineni

GenAI / Applied AI Engineer

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Work authorization: German permanent settlement permit (Niederlassungserlaubnis)



Summary

GenAI and Applied AI Engineer with 7+ years of experience building machine learning, deep learning, and generative AI systems from scratch through production. As Founding AI Lead at an EXIST-funded startup, delivered production-grade LLM, RAG, and agentic systems on Google Cloud and Azure, including German clinical AI solutions. Proven ability to translate research into scalable, reliable systems.

Core Skills

- **Generative AI and LLMs:** RAG, AI agents and orchestration (LangChain, LangGraph), LLM fine-tuning, prompt engineering, vector databases (Qdrant, FAISS, Azure AI Search, Chroma)
- **Machine learning and deep learning:** computer vision (image classification, object detection, instance segmentation), NLP, multimodal and vision-language models, classical ML (Decision Tree, Random Forest, XGBoost)
- **Cloud and MLOps:** Google Cloud (Vertex AI), Microsoft Azure (Azure AI), AWS, CI/CD, HPC and GPU clusters, containerized deployment
- **Programming and frameworks:** Python (PyTorch, TensorFlow, Keras, scikit-learn, NumPy, pandas); LangChain, LangGraph; MATLAB

Experience

Founding AI Lead, Nixi AI

May 2025 – May 2026

Wiesbaden, Germany / Remote. EXIST-funded medical AI startup (backed by the German Federal Ministry for Economic Affairs and Energy).

- Built and shipped production generative-AI systems on Google Cloud (Vertex AI) and Microsoft Azure, from data pipelines through to deployed inference services and APIs.
- Built retrieval-augmented generation (RAG) pipelines for clinical-evidence synthesis using vector databases (Qdrant, FAISS, Azure AI Search).
- Orchestrated agentic, multi-step LLM workflows (LangGraph, LangChain) for clinical decision-support and documentation.
- Developed German medical speech-to-text (ASR) fine-tuning, automated ICD-10-GM coding, and GDPR-compliant clinical-text anonymization.
- Devised AI engineering from the ground up; co-authored grant proposals, supported hiring, and supervised thesis students.

AI Engineer / PhD Researcher, Friedrich-Schiller-Universität Jena

Nov 2020 – Present

Jena, Germany. Heinz Nixdorf Chair for Distributed Information Systems (funded contract Nov 2020 – Dec 2024). Application domain: biodiversity.

- Developed end-to-end deep learning pipelines for image classification, object detection, and instance segmentation on a custom image dataset developed in-house.
- Designed a RAG-supported LLM pipeline to extract methods and findings from hundreds of research articles.
- Built a pipeline that automatically constructs knowledge graphs from unstructured text using LLMs.
- Integrated multimodal vision-language models to extract information from images via natural-language prompts.
- Co-taught a graduate seminar (2024); supervised Master's thesis students and data annotation teams.
- Presented at international conferences (SEMANTiCS, TDWG, GfÖ) and organized scientific sessions (TDWG, GfÖ).

Machine Learning Engineer (Student Assistant) and M.Sc. Researcher, Max Planck Institute for Biogeochemistry

Apr 2019 – Sep 2020

Jena, Germany.

- Built an end-to-end machine learning project covering data collection, annotation, training, feature engineering, and evaluation.
- Developed classical ML regression models (Decision Tree, Random Forest, XGBoost) to predict 14,721 global leaf-area values.
- Ran annotation projects for image classification, object detection, and segmentation.

Intern, Signal Processing, University of Applied Sciences Jena

Apr 2018 – Aug 2018

Jena, Germany.

- Reconstructed noisy signals using compressed sensing and Fourier transforms across varying noise thresholds, then assessed signal integrity.

Education

PhD Candidate, Computer Science, Friedrich-Schiller-Universität Jena

Nov 2020 – Present

(expected 2026)

Thesis: A Quality-Aware, Provenance-Complete, Interoperable Multimodal Knowledge Graph for Digital Herbaria Images.

M.Sc., Scientific Instrumentation, University of Applied Sciences Jena (EAH Jena)

Oct 2016 –

Jul 2020

Thesis: Evaluating global patterns of leaf areas using machine learning approaches.

Selected Publications

- Kommineni VK, König-Ries B, Samuel S (2025). Multi-LLM information retrieval pipeline for extracting deep learning methodologies in biodiversity research. PeerJ Computer Science 11:e3204.
- Ahmed W, Kommineni VK, König-Ries B, Gaikwad J, Gadelha L, Samuel S (2025). Evaluating the method reproducibility of deep learning models in biodiversity research. PeerJ Computer Science 11:e2618.
- Kommineni VK, König-Ries B, Samuel S (2024). Towards the Automation of Knowledge Graph Construction using Large Language Models. NLP4KGC at SEMANTICS 2024 (CEUR-WS Vol-3874).

Full list: scholar.google.com/citations?user=ZEz2nk4AAAAJ

Languages

English (business fluent) | German (B1 certified, telc; B2 in progress) | Telugu (native)